

# Hash Prototype Personalized Federated Learning for Multi-Site Brain Network Analysis

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**Abstract**—Multi-site brain network analysis methods have garnered significant attention due to their ability to leverage diverse samples from multiple hospitals. Meanwhile, the growing emphasis on medical privacy protection has driven the development of secure multi-site learning approaches, such as federated learning. However, the challenges of cross-site data distribution alignment and personalized model construction under privacy constraints remain inadequately addressed, which limits the effectiveness of existing multi-site federated approaches. In this paper, we propose a hash prototype-based personalized federated learning method for multi-site brain network analysis, which enables effective cross-site data distribution calibration and neurological pattern recognition. Firstly, we decouple local models into shared encoders to preserve site-specific features and a hash encoding layer to generate prototype representations with reduced information exposure. Secondly, these hash prototypes are aggregated and disseminated across sites, facilitating the generation of prototypes that guide cross-site data distribution alignment. Additionally, encoder parameters from different sites are dynamically aggregated, fusing global and local knowledge while preserving the personalized feature extraction capabilities of each site. Finally, we validate the personalized models across multiple sites. Extensive experiments on Autism Brain Imaging Data Exchange (ABIDE), Attention Deficit Hyperactivity Disorder (ADHD), and Schizophrenia (SZ) datasets demonstrate that our method outperforms existing state-of-the-art multi-site methods. Furthermore, our method can more effectively identify multi-site brain network biomarkers due to the learned prototypes.

**Index Terms**—Federated learning, multi-site, brain network analysis, hash prototype, neurological pattern recognition

## I. INTRODUCTION

NEUROIMAGING analysis has become an essential tool for understanding complex brain functions, characterizing underlying pathological mechanisms, and supporting the understanding of neuropsychiatric patterns [1], [2]. Compared with single-site approaches, multi-site collaborative diagnostic paradigms for brain diseases integrate heterogeneous datasets from diverse medical institutions, significantly enhancing the

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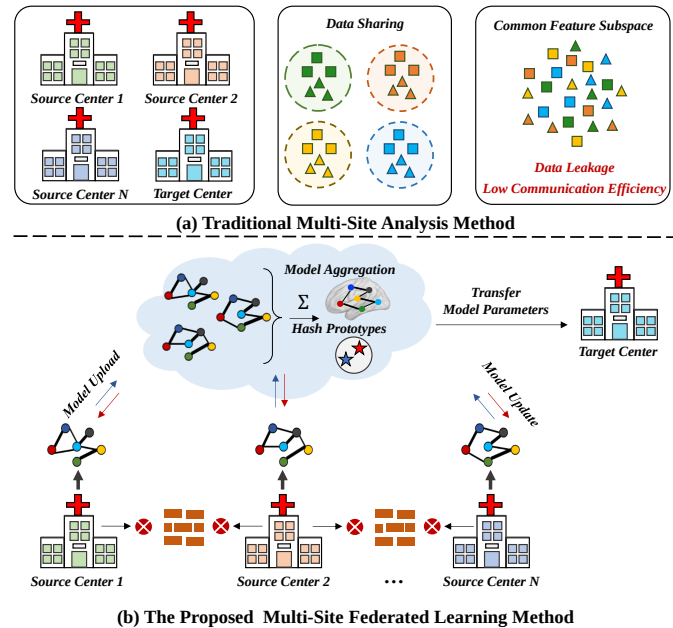


Fig. 1. The schematic diagram of the multi-site federated learning method.

robustness of brain disease diagnosis [3]. However, the cross-site sharing of sensitive medical data raises severe privacy concerns. This conflict between the demand for large-scale data collaboration and the necessity of privacy preservation highlights the critical importance of developing privacy-preserving multi-site models for brain network analysis [4], [5]. Therefore, addressing this challenge is essential to balance advances in medical knowledge with the protection of patient privacy.

The emergence of federated learning provides a potential solution to the data security challenges inherent in multi-site collaborative diagnosis [6], [7]. As a decentralized learning paradigm, federated learning eliminates the need to centralize raw data, thereby mitigating privacy risks by enabling model training at local sites and exchanging only aggregated model parameters. Zhang *et al.* [8] proposed a federated learning framework designed to train deep neural networks by leveraging the local data characteristics of each client. Mao *et al.* [9] introduced federated and split learning techniques to decompose a spatio-temporal model into a client-side temporal model and a server-side spatial model. Peng *et al.* [10] proposed a federated knowledge-enhanced initialization framework, which leverages cross-client and cross-task transfer of prior knowledge to generate informed initializations for

learning new tasks with adapters. However, existing federated learning methods are susceptible to catastrophic forgetting during training and show notable limitations in aligning heterogeneous data distributions across sites.

Many studies have shown that the transmission of model parameters in federated learning still introduces privacy leakage risks [11], [12]. Liu *et al.* [12] introduced a method that applies local differential privacy to perturb model parameters during transmission, thereby reducing the risk of privacy leakage. Li *et al.* [13] proposed a multi-stage adaptive privacy-preserving algorithm to improve the trade-off between model utility and privacy in differentially private asynchronous federated learning. However, injecting random noise often alters the topological connections of functional magnetic resonance imaging (fMRI) brain networks and degrades the feature extraction capability, leading to a significant drop in diagnostic accuracy across heterogeneous sites. Therefore, explicitly protecting privacy without destroying the deterministic structural semantics remains a critical challenge.

The discovery of reliable biomarkers is a critical task in multi-site brain disease analysis [14]–[16]. The selection of discriminative biomarkers and representative samples remains a major challenge in achieving robust diagnostic performance. Ren *et al.* [17] proposed Brain-Shapelet, a framework that extracts discriminative subsequences from fMRI data to identify potential instantaneous biomarkers of brain activity. Song *et al.* [18] integrated a Transformer module into a graph convolutional network classifier to model relationships between functional connectivity network features, aiming to derive more discriminative neuroimaging biomarkers. Yang *et al.* [19] introduced the BrainMass model, which can identify patterns of various brain disorders and pinpoint key biomarkers. Although existing studies have explored this direction, they either suffer from information leakage or rely on federated learning methods that focus solely on site-specific biomarkers, thereby limiting their reliability.

In order to address the above problems, we propose an innovative hash prototype-based personalized federated learning method for multi-site brain network analysis. Firstly, we decouple the local models into shared encoders and private decoders, and introduce a hash encoding layer to generate compact binary hash representations with reduced information exposure. Secondly, after local training, each site transmits its hash prototypes to the server instead of sharing raw data, thereby reducing direct exposure of raw neuroimaging data. The server performs class-wise federated averaging over local prototypes to obtain global class-level prototypes, which are then broadcast back to all sites and used as references for prototype-guided feature alignment. To enable personalization during model synchronization, we employ a learnable dynamic aggregation strategy that facilitates the integration of global knowledge and local specificity for each site's encoder. Finally, we use a multi-layer perceptron (MLP) approach to classify the extracted features for multi-site brain network analysis. The comparisons of our method and the existing multi-site analysis method are shown in Fig. 1. Fig. 1 (a) shows the conventional multi-site analysis method faces several limitations. Strict privacy regulations and data-sharing restrictions hinder the direct

exchange of raw medical data, limiting large-scale integration. Moreover, conventional federated learning approaches often build a unified model without considering site-specific or individual characteristics, resulting in limited personalization. Fig. 1 (b) presents our multi-site hash prototype personalized federated learning method, which enables secure cross-site collaboration while simultaneously capturing both global and site-specific biomarkers to improve diagnostic performance. In brief, the main contributions of our method are summarized as follows:

This paper presents a hash-prototype-driven personalized federated learning framework tailored for multi-site brain network analysis, which not only achieves effective cross-site data distribution alignment but also avoids the need for medical data transmission between centers.

1) We propose a hash federated prototype generation method that integrates local and global knowledge, allowing each site to construct compact binary hash prototypes for cross-site distribution alignment without sharing raw neuroimaging data.

2) We develop a dynamic aggregation strategy that updates the local shared encoder through adaptive integration with the server-side federated encoder, thereby preserving the site-specific feature extraction capability while benefiting from global collaboration.

3) Extensive experiments conducted on three datasets demonstrate that our proposed method outperforms several state-of-the-art approaches in multi-site brain network analysis. These results highlight the effectiveness of our framework in identifying multi-site brain network biomarkers.

The remainder of this paper is organized as follows: Section II provides an elaboration on the related work; Section III presents the proposed model in detail; Section IV outlines the experimental setup and reports the results; Section V discusses the key findings and implications of the experiments; Finally, Section VI concludes the paper.

## II. RELATED WORK

### A. Federated Learning

Federated learning has emerged as an effective paradigm for collaborative model training across multiple sites while preserving data privacy, which is essential in healthcare settings [20], [21]. The standard federated learning process involves initializing a global model  $\theta_{global}^t$  on a central server, which is distributed to each site at communication round  $t$ . Each local site updates this model based on its private dataset using the update rule:

$$\theta_{local}^{t+1} = \theta_{local}^t - \eta \nabla L(\theta_{local}^t) \quad (1)$$

where  $L$  denotes the local loss function at a specific site,  $\eta$  is the learning rate,  $\nabla$  represents the gradient operator, and  $t$  represents the communication round. The updated local models are then averaged to update the global model:

$$\theta_{global}^{t+1} = \frac{1}{M} \sum_{i=1}^M \theta_{local,i}^{t+1} \quad (2)$$

where  $i$  denotes the  $i$ -th site, and  $M$  represents the number of participating sites.

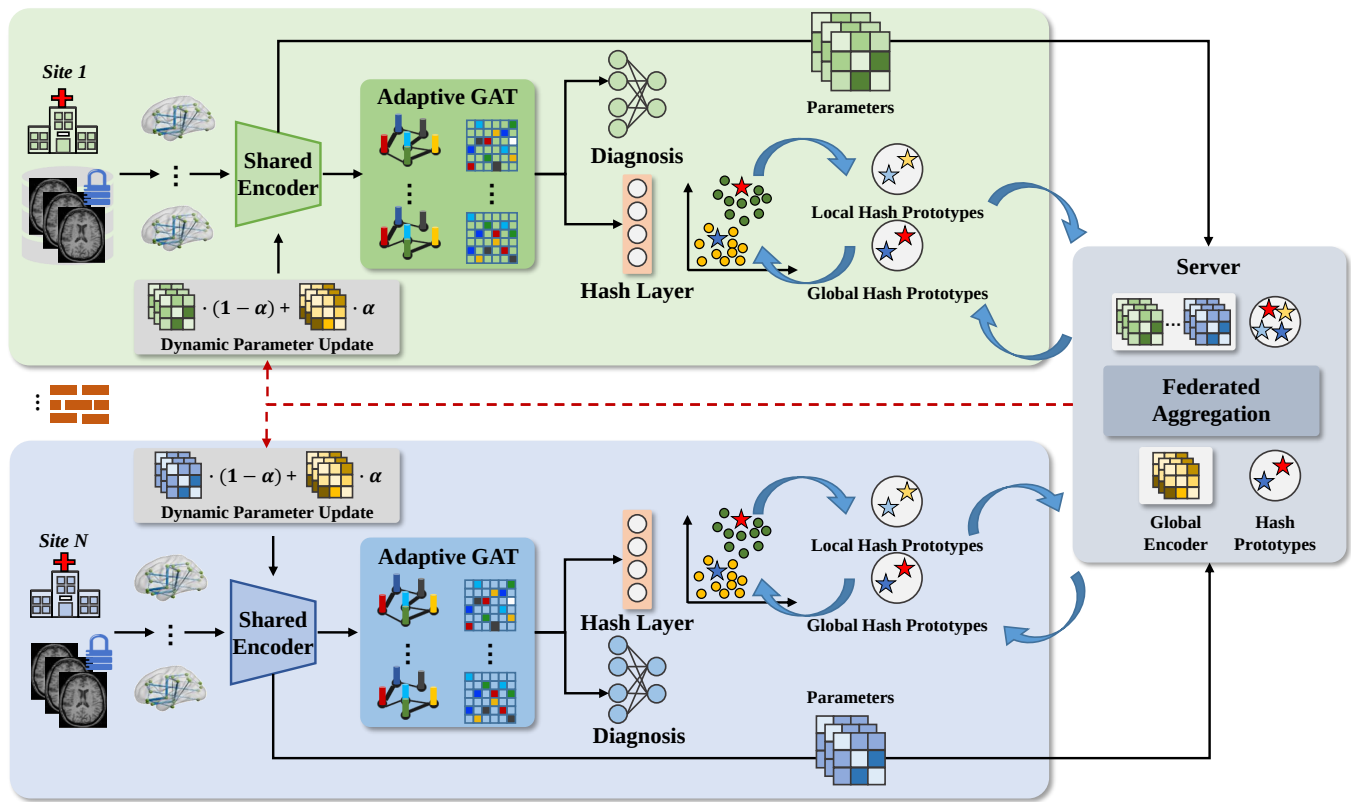


Fig. 2. The framework of the proposed hash prototype personalized federated learning method for multi-site brain network analysis.

Personalized federated learning is developed to adapt shared models to the distinct data distributions of individual clients, thereby enhancing model relevance while ensuring privacy preservation and computational efficiency. Methods such as FedBN [22] employ local batch normalization to mitigate feature shifts caused by heterogeneous data distributions across sites. Another personalized federated learning strategy, FedRep [23], learns a shared global representation while enabling local adaptation, thereby facilitating efficient convergence with reduced sample complexity. FedProto adopts a prototype-based communication strategy by sharing class prototypes in a common feature space, followed by prototype aggregation across clients [24]. However, existing personalized federated learning methods still face significant challenges in aligning cross-site data distributions and reducing the risk of information leakage during cross-site transmission.

### B. Deep Hash Retrieval

Deep hash retrieval leverages neural networks to convert complex data features into compact binary codes, facilitating efficient data retrieval across extensive databases [25], [26]. Cao *et al.* [27] introduced HashNet, which applies a neural network for a nonlinear transformation of input data into a Hamming embedding space, where samples from the same class exhibit small Hamming distances, thereby improving discriminability. HashNet adopts a continuation strategy to transition from smooth activations to a sign function, generating binary representations that preserve semantic information. Deep hashing methods further enhance privacy and

communication efficiency by transforming data into compact and less interpretable codes, making them suitable for federated learning. For example, FedHAP employs hashing to reduce output sizes and enhance federated learning efficiency through a binary cross-entropy-based quantization loss, which simultaneously increases accuracy and reduces communication costs [28]. However, FedHAP relies on CNN-based extractors for image data, focusing on pixel-level spatial correlations. Consequently, it lacks the capability to model the irregular topological structures and dynamic functional connectivity inherent in brain networks.

### III. PROPOSED METHOD

### A. Overview

In this section, we provide a detailed introduction to the proposed method for multi-site brain network analysis, and its schematic diagram is depicted in Fig. 2. The training process follows an iterative procedure, including local feature learning, prototype generation, cross-site prototype interaction, and personalized model aggregation. The shared encoder with adaptive GAT takes the local brain network adjacency matrices as input and utilizes an adaptive GAT to capture complex topological relationships among brain regions. It dynamically assigns attention weights based on patient-specific connections and outputs continuous graph feature representations. The hash prototype module applies a sign function to map the continuous features into a discrete Hamming space, generating binary hash codes. Subsequently, it calculates the mode of the

hash codes for each class to generate local hash prototypes. Crucially, the exchange of hash prototypes effectively reduces information exposure, allowing for collaborative multi-site training without sharing raw medical data. To preserve the site-specific clinical data distribution, the private decoder is updated strictly locally and is never transmitted to the central server. Each local site downloads the global model and employs a learnable gating mechanism to dynamically fuse global knowledge with local parameters. This step ensures a balance between generalized knowledge and personalized diagnostic capability.

### B. Local Training

1) *Feature Extraction*: In the proposed federated framework, each site performs local training to learn representations from its own brain imaging data while preserving data privacy. This process aims to capture site-specific features and contribute to the global model without directly sharing raw data. To model the intrinsic relationships among brain regions, the local data are first transformed into a graph structure. Graph-based representations have demonstrated remarkable efficacy in modeling topological relationships [29]. The adjacency matrix represents the connections between different brain regions. Within the graph attention network (GAT) [30], connections are adaptively refined through attention coefficients  $\alpha_{ij}$  learned during training, thereby enabling the model to adaptively emphasize more informative inter-regional relationships [30].

Feature extraction is carried out using a shared encoder and an adaptive GAT layer to generate the diagnostic feature  $H$  and topological feature  $A$  of the brain network. The shared encoder allows for a consistent representation of the brain imaging data across different sites. By sharing the same encoder architecture and parameters, the model can learn a generalized feature representation that is robust across different datasets. The parameters of the shared encoder  $W_{encoder}$  are obtained through a federated aggregation process described in the Federated Aggregation section.

To accommodate distributional differences among sites, a learnable gating mechanism is introduced to adaptively fuse global and local encoder parameters. The gating weights are optimized during training, allowing each site to dynamically balance global knowledge and local specificity. This adaptive fusion enhances both generalization and personalization, ensuring that the model captures site-specific characteristics while remaining aligned with the global representation. To further improve training stability and convergence, batch normalization is applied to the output features of each layer. Specifically, for the feature representation  $H^{l+1}$  at layer  $l+1$ , normalization is defined as:

$$H_{norm} = \frac{H^{l+1} - \mu}{\sqrt{\sigma^2 + \epsilon}} \quad (3)$$

where  $\mu$  and  $\sigma^2$  are the mean and variance of the features, and  $\epsilon$  is a small constant to avoid division by zero. Normalization helps in stabilizing and accelerating the training process by maintaining the scale of the features and reducing internal covariate shift.

2) *Hash Encoding*: To reduce information leakage during sharing, the learned features are further transformed into compact binary codes through a hash encoding module. Specifically, the topological feature  $A$  is passed through a hash layer to generate hash codes  $H_{hash}$ :

$$H_{hash} = \text{sign}(Z_{hash}) = \text{sign}(AW_{hash} + b_{hash}) \quad (4)$$

where the sign function is employed to binarize the continuous representations  $Z_{hash}$  into a discrete hash code,  $W_{hash}$  and  $b_{hash}$  represent the weights and bias vector of the hash layer. If the hash prototype is intercepted, reconstructing the original neural network from the binary codes becomes a challenging problem, effectively reducing information exposure. Then, we calculate quantization loss to ensure the accuracy of preserving the original data structure within the binary code space:

$$\mathcal{L}_Q = \frac{1}{N} \sum_{i=1}^N \| |Z_{hash}^i| - \mathbf{1} \|_1 \quad (5)$$

where  $\mathbf{1}$  is an all-one vector of the same dimension as the hash codes. Additionally, the hash codes are used to calculate the distance  $\mathcal{D}$  to the hash prototype  $P^c$ , yielding the prototype distance loss  $\mathcal{L}_{PD}$ . The prototype distance loss ensures that the hash-coded adjacency matrices remain close to their corresponding hash prototypes:

$$\mathcal{D}(H_{hash}^c, P^c) = \| H_{hash}^c - P^c \|_2^2 \quad (6)$$

where  $\| \cdot \|_2^2$  denotes the squared Frobenius norm. The prototype distance loss is defined as:

$$\mathcal{L}_{PD} = \frac{1}{|H_{hash}|} \sum_{c=1}^K \mathcal{D}(H_{hash}^c, P^c) \quad (7)$$

where  $K$  is the number of classes.

In the first training round, hash prototypes at each site are initialized using bit-wise majority voting over the hash codes of samples within the same class. This ensures that the initial hash prototypes represent the dominant characteristics of each class. These initialized prototypes serve as references for subsequent model training. In subsequent training rounds, hash prototypes are updated to maintain sufficient separation in the Hamming space. Inspired by the Gilbert–Varshamov bound from coding theory, a minimum Hamming distance is enforced between class prototypes to improve inter-class discriminability. The steps are as follows:

(1) Define the number  $c$  and the length  $q$  of hash codes: The number of hash prototypes  $c$  is determined by the number of classes at each site, while the hash length  $q$  is predefined according to system requirements.

(2) Calculate the minimal Hamming distance  $d$ : The Gilbert–Varshamov bound ensures that for  $c$  binary codes of length  $q$ , there exists a minimal Hamming distance  $d$  between any two codes, provided that:

$$2^q \geq c \cdot \sum_{i=0}^{d-1} \binom{q}{i} \quad (8)$$

This inequality determines the maximum  $d$  such that the sum of the binomial coefficients from 0 to  $d-1$  does not exceed  $2^q/c$ .

**Algorithm 1** Training procedure of our proposed federated learning method.

**Input:** Local datasets at each site  $\{D_i\}_{i=1}^M$ , where  $M$  is the number of sites.

**Output:** Each local model parameters  $W_{\text{encoder}}^i$ , updated hash prototypes  $\{P_c^i\}_{c=1}^K$ , where  $K$  is the number of classes.

```

1: Initialize shared encoder parameters  $W_{\text{encoder}}$ , hash prototypes  $\{P_c^i\}_{i=1}^M$  for each class  $c$  at each site  $i$ 
2: for each training round  $t$  do
3:   for each site  $i$  do
4:      $H \leftarrow \text{shared\_encoder}(D_i, W_{\text{encoder}})$  // Extract features
5:      $H_{\text{norm}} \leftarrow \text{BatchNorm}(H)$  // Normalize features
6:      $H_{\text{pooled}} \leftarrow \text{add\_pooling}(H_{\text{norm}})$  // Pool features
7:      $E \leftarrow \text{fully\_connected}(H_{\text{pooled}})$  // Transform into embedding space
8:      $H_{\text{hash}} \leftarrow \text{hash\_layer}(A)$  // Map topological feature into Hamming space
9:      $z \leftarrow \text{private\_decoder}(E)$  // Perform binary classification
10:    Calculate the cross-entropy loss  $\mathcal{L}_{CE}$  by Eq. (12)
11:    The hash feature  $H_{\text{hash}}$  is used to calculate the distance  $\mathcal{D}$  to the corresponding hash prototype  $P_c$  by Eq. (6)
12:    Calculate the prototype distance loss  $\mathcal{L}_{PD}$  by Eq. (7)
13:    Calculate the quantization loss  $\mathcal{L}_Q$  by Eq. (5)
14:    Calculate the combined loss  $\mathcal{L}$  by Eq. (13)
15:    Update  $W_{\text{encoder}}$ ,  $P_c^i$ , and gating weight  $\alpha$  based on  $\mathcal{L}$  // Update model parameters locally
16:  end for
17:  Collect  $\{W_{\text{encoder}}^i\}_{i=1}^M, \{P_c^i\}_{i=1}^M$  at central server
18:   $W_{\text{global}} \leftarrow \frac{1}{M} \sum_{i=1}^M W_{\text{encoder}}^i$  // Federated aggregation of encoder
19:  for each site  $i$  do
20:     $W_{\text{updated}} \leftarrow \alpha W_{\text{global}} + (1-\alpha)W_{\text{encoder}}^i$  // Fuse global and local parameters dynamically
21:  for each class  $c$  do
22:     $P_{\text{new}}^c \leftarrow \frac{1}{M} \sum_{i=1}^M P_c^i$  // Aggregate class-wise hash prototypes by averaging
23:  end for
24: end for
25: end for

```

(3) Generate initial hash codes: To generate  $c$  hash codes of length  $q$  with a minimal distance  $d$ , an exhaustive search is performed to find the maximum  $d$  that satisfies the Gilbert-Varshamov bound. The following iterative process can be used to generate the hash codes:

$$d^* = \max \left\{ d : 2^q \geq \sum_{i=0}^{d-1} \binom{q}{i} \cdot c \right\} \quad (9)$$

(4) Implementation of hash centers: Each new hash prototype is assigned to a specific class. These updated hash centers ensure that images from different classes remain distinctly sep-

arated in the Hamming space for subsequent training rounds.

3) *Multi-Site Classification Task*: Based on the learned feature representations, the model performs downstream brain disease diagnosis. Specifically, the normalized node features are aggregated to obtain a compact graph-level representation. Perform add-pooling on the normalized features to reduce dimensionality:

$$H_{\text{pooled}} = \sum_{i \in V} H_{\text{norm}} \quad (10)$$

where  $V$  represents the set of brain region nodes. A fully connected layer is then used to transform the pooled features into the embedding space  $E$ :

$$E = \text{ReLU}(H_{\text{pooled}}W_{fc} + b_{fc}) \quad (11)$$

where  $W_{fc}$  and  $b_{fc}$  denote the weight and bias parameters of the fully connected layer, respectively.

The resulting embeddings are fed into a private decoder (implemented as a multilayer perceptron, MLP) for binary classification. The private decoder is trained by minimizing the cross-entropy loss  $\mathcal{L}_{CE}$ :

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{y}_i + (1 - y_i) \log (1 - \hat{y}_i)] \quad (12)$$

where  $y_i$  denotes the true label and  $\hat{y}_i$  represents the predicted probability for the  $i$ -th sample.

Combine the cross-entropy loss, the prototype distance loss, and the quantization loss to form the total loss for training:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda_1 \mathcal{L}_{PD} + \lambda_2 \mathcal{L}_Q \quad (13)$$

where  $\lambda_1$  and  $\lambda_2$  are balancing hyperparameters,  $\mathcal{L}_{CE}$  represents the label classification loss,  $\mathcal{L}_{PD}$  denotes prototype distance loss, and  $\mathcal{L}_Q$  represents quantization loss.

### C. Federated Aggregation

The central server collects all hash prototypes from the sites. The hash prototypes of the same class from multiple sites are aggregated by directly averaging them to produce new prototypes:

$$P_{\text{new}}^c = \frac{1}{M} \sum_{i=1}^M P_c^i \quad (14)$$

where  $P_c^i$  denotes the hash prototype of class  $c$  from the  $i$ -th site, and  $M$  is the number of sites. For each class, the server collects the local hash prototypes from the participating sites and computes their average to obtain the global hash prototype. The resulting prototype is then broadcast to all sites and used as a global class-level reference in the prototype distance loss during the next local training round.

The aggregated parameters  $W_{\text{global}}$  are then transmitted back to each site for the next round of local training, enabling iterative global-local adaptation under the federated learning framework. Each site fuses the aggregated parameters with its original shared encoder parameters using a gating operation. The global and local parameters are dynamically integrated

during the training process. The learnable gating weights  $\alpha$  are updated through backpropagation, allowing the model to adapt the fusion process over time. Dynamic parameter update module maintains global consistency while adapting to the local characteristics of each site, enhancing the model's generalization ability. The fusion of global and local parameters is performed via a dynamic parameter update strategy, which adaptively adjusts the model weights during training.

$$W_{updated} = \alpha W_{global} + (1 - \alpha) W_{encoder}^i \quad (15)$$

where  $\alpha$  is a learnable gating weight that is updated during training via backpropagation.

The detailed steps of the training optimization process, including model parameter updates and communication between clients and the central server, are presented in Algorithm 1.

#### IV. EXPERIMENT

##### A. Datasets and Preprocessing

We evaluated our method on three brain disease datasets: Autism Brain Imaging Data Exchange (ABIDE), Attention Deficit Hyperactivity Disorder (ADHD), and Schizophrenia (SZ) [31]–[33]. The ABIDE dataset includes resting-state fMRI (rs-fMRI), anatomical, and phenotypic data collected from 17 international sites and made publicly available. We selected the five sites with the largest sample sizes, which constituted more than half of the entire dataset, and excluded the sites with insufficient data. The ADHD-200 dataset is a multi-site dataset comprising eight international imaging sites, including neuroimaging and phenotypic data from 362 children and adolescents with ADHD and 585 healthy controls. Three sites were excluded from our analysis due to missing data labels or severe class imbalance, leaving five sites for further study. Since our model training assumed independent and non-overlapping datasets across sites, we observed substantial class imbalance among the three ADHD subtypes. Consequently, we restricted the classification task to distinguishing between ADHD and healthy controls, rather than further differentiating subtypes. The SZ dataset consists of rs-fMRI data from five sites: Nanjing Brain Hospital (NBH), the Center for Biomedical Research Excellence (COBRE), Nottingham, Taiwan, and Xiangya. The subjects satisfied the following criteria: (1) no other DSM-IV disorder, (2) no history of drug abuse, (3) no clinically significant head trauma.

For model development, we preprocessed the rs-fMRI data using the Automated Anatomical Labeling (AAL) template [34], which partitions the brain into regions of interest (ROIs) based on anatomical criteria. AAL is widely used in neuroimaging research as a standard atlas for analyzing brain activation and connectivity across subjects and modalities. For each subject, we extracted the mean time series from each ROI, resulting in a matrix of size  $N \times R$ , where  $N$  is the number of time points, and  $R$  is the number of ROIs. The functional connectivity matrix was then computed by calculating the Pearson correlation coefficient between every pair of ROIs, using the following formula:

$$r_{xy} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2 \sum_{i=1}^N (y_i - \bar{y})^2}} \quad (16)$$

where  $x_i$  and  $y_i$  denote the time series values of ROIs  $x$  and  $y$  at the  $i$ -th time point, and  $\bar{x}$  and  $\bar{y}$  denote their temporal means. We obtain an  $R \times R$  correlation matrix, with each element denoting the correlation coefficient between two ROIs. Accordingly, this matrix is transformed into a graph by treating the ROIs as nodes and the coefficients as edge weights, thus establishing the graph structure. This graph representation serves as the input to the graph attention network for modeling brain network topology.

In our experimentation, our framework was implemented with PyTorch, and all experiments were performed on one NVIDIA GeForce RTX 3080 GPU. For each site, we validated our method and all comparative methods using five-fold cross-validation. For each cross-validation fold, each site uses one-fold as the local test set, while the remaining data at that site are used for model training and validation. The validation set used for hyperparameter selection is separated from the corresponding test fold and is never used for model evaluation. The results are reported with the corresponding standard deviations across the five folds. The Adam optimizer is employed for parameter updates, with the learning rate set to 0.05 and the batch size set to 16. In our experiments, we set the number of federated learning rounds to 100 and the number of local training epochs per round to 3. In our implementation, the hyperparameters  $\lambda_1$  and  $\lambda_2$  are systematically varied within the range  $\{0.1, 0.2, 0.3, \dots, 1\}$ . A grid search strategy is employed on a validation set to determine the optimal combination by maximizing the validation Top-1 accuracy. On the ABIDE dataset, the best performance is achieved with  $\lambda_1 = 0.3$  and  $\lambda_2 = 0.3$ . During the training process, the learnable gating weight  $\alpha$  is uniformly initialized to 0.5 for all participating sites.

$$\alpha = \text{sigmoid}(\text{alpha\_logit}) \quad (17)$$

$\text{alpha\_logit}$  is defined as a learnable parameter and is optimized jointly with the other trainable model parameters via backpropagation using the Adam optimizer. Since the actual fusion coefficient is computed as  $\text{sigmoid}(\text{alpha\_logit})$ , the resulting  $\alpha$  is always constrained to  $[0, 1]$ , which makes the fusion process stable.

##### B. Competing Methods

We conduct a comparative analysis between the proposed method and several federated learning approaches, including FedAvg [35], FedProx [36], MOON [37], FedBN [22], FedRep [23], Ditto [38], FedBABU [39], FedProto [24], CusFed [40], FedALA [41], FedPAC [42], and FedHAP [28]. For all methods, we adopt a unified architecture, where a shared encoder serves as the feature extractor and a private decoder is used as the classifier.

- **FedAvg:** FedAvg is a fundamental baseline for federated learning, where client devices update a global model by exchanging their local model updates.
- **FedProx:** FedProx is an extension of FedAvg that introduces proximal terms to promote faster convergence and robustness. By incorporating local proximal terms during

TABLE I  
THE CLASSIFICATION RESULTS±STD (%) OF THE PROPOSED METHOD AND DIFFERENT FEDERATED METHODS ON ABIDE, ADHD-200, AND SCHIZOPHRENIA DATASETS. "\*\*\*" INDICATES A SIGNIFICANT IMPROVEMENT ( $p < 0.05$ ).

| Methods       | ABIDE       |             |             | ADHD-200    |             |             | Schizophrenia |             |             |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------|-------------|-------------|
|               | ACC         | F1          | AUC         | ACC         | F1          | AUC         | ACC           | F1          | AUC         |
| Local only    | 65.65±0.10  | 60.18±0.33  | 58.10±0.09  | 66.58±0.05  | 39.37±0.09  | 58.10±0.05  | 68.44±0.04    | 44.06±0.14  | 56.93±0.05  |
| FedAvg [35]   | 70.10±0.03  | 54.42±0.14  | 58.73±0.09  | 66.41±0.06  | 46.46±0.01  | 56.06±0.07  | 69.18±0.03    | 52.08±0.05  | 55.51±0.08  |
| FedProx [36]  | 68.25±0.08  | 56.22±0.14  | 70.38±0.09  | 68.95±0.13  | 52.93±0.34  | 60.86±0.02  | 69.85±0.09    | 51.64±0.02  | 57.32±0.01  |
| MOON [37]     | 72.59±0.02  | 59.32±0.32  | 63.37±0.12  | 69.29±0.13  | 44.20±0.05  | 60.44±0.10  | 69.66±0.10    | 45.88±0.07  | 58.40±0.05  |
| FedBN [22]    | 70.99±0.07  | 60.82±0.11  | 66.97±0.10  | 71.16±0.06  | 50.39±1.29  | 63.54±0.09  | 72.40±0.10    | 50.63±1.33  | 62.81±0.02  |
| FedRep [23]   | 72.61±0.11  | 56.65±0.02  | 67.85±0.11  | 69.64±0.12  | 47.32±0.70  | 60.28±0.11  | 71.46±0.06    | 45.17±0.66  | 63.48±0.15  |
| Ditto [38]    | 69.85±0.05  | 60.07±0.15  | 65.98±0.12  | 71.71±0.09  | 57.32±0.06  | 64.72±0.12  | 71.23±0.10    | 47.88±0.37  | 66.21±0.02  |
| FedBABU [39]  | 72.67±0.07  | 64.63±0.29  | 62.31±0.06  | 70.97±0.02  | 56.24±0.14  | 60.40±0.15  | 69.94±0.02    | 57.21±0.07  | 69.04±0.02  |
| FedProto [24] | 73.51±0.04  | 55.86±0.45  | 57.25±0.04  | 70.66±0.08  | 59.45±0.32  | 64.60±0.11  | 71.55±0.14    | 42.07±0.06  | 57.95±0.09  |
| CusFed [40]   | 75.12±0.04* | 60.45±0.58* | 67.21±0.09* | 70.73±0.11  | 53.59±0.45  | 61.34±0.04  | 72.27±0.07    | 59.51±0.03  | 67.58±0.07  |
| FedALA [41]   | 73.01±0.16  | 62.51±0.07  | 69.93±0.04  | 72.89±0.01* | 55.91±0.46* | 56.61±0.06* | 72.24±0.02    | 57.85±0.38  | 56.88±0.02  |
| FedPAC [42]   | 71.43±0.03  | 64.41±0.26  | 66.37±0.09  | 72.26±0.18  | 58.19±0.09  | 66.94±0.06  | 70.86±0.54    | 63.92±0.18  | 63.27±0.26  |
| FedHAP [28]   | 73.11±0.04  | 54.99±0.40  | 59.64±0.09  | 70.89±0.13  | 48.53±1.30  | 66.99±0.05  | 72.59±0.12*   | 43.94±0.09* | 61.05±0.04* |
| Ours          | 78.78±0.09  | 62.22±0.21  | 74.58±0.08  | 76.84±0.31  | 65.77±0.13  | 75.32±0.29  | 77.77±0.34    | 71.18±0.52  | 73.31±0.32  |

the aggregation process, FedProx addresses the issue of stragglers and non-IID data distribution.

- **MOON:** MOON (model-contrastive learning) corrects the local updates by maximizing the agreement of the representations learned by the current local model and the global model.
- **FedBN:** FedBN is also an extension of FedAvg. It differs from FedAvg in that it ignores the parameters of the batch normalization layers during the parameter aggregation process.
- **FedRep:** FedRep efficiently learns a shared data representation across clients with unique local heads. It achieves linear convergence to the ground-truth representation and demonstrates superior performance in heterogeneous settings compared to alternative personalized federated learning approaches.
- **Ditto:** Ditto aims to address the fairness and robustness issues in federated learning. It achieves this by sharing model parameters among devices and optimizing them using a global objective function.
- **FedBABU:** FedBABU addresses the personalization performance degradation by updating only the body of the model during federated training. The head is fine-tuned for personalization during the evaluation process, resulting in consistent performance improvements and efficient personalization.
- **FedProto:** FedProto is a novel framework for federated learning that enhances tolerance to client heterogeneity. Instead of communicating gradients, it exchanges abstract class prototypes. This approach leads to better convergence and performance on multiple datasets in a benchmark setting tailored for heterogeneous federated learning.
- **CusFed:** CusFed is a novel federated method that combines the contrastive learning algorithm to learn latent representations, leading to improved performance in med-

ical image analysis tasks.

- **FedALA:** FedALA addresses the statistical heterogeneity in federated learning by utilizing an adaptive local aggregation module. This module adaptively aggregates the global and local models for personalized client models, thereby improving generalization and initialization during training iterations.
- **FedPAC:** FedPAC is a federated learning framework that aligns features and combines classifiers for fast and high performance. It uses a shared space to reduce data differences and a collaboration module to optimize and weight classifiers.
- **FedHAP:** FedHAP applies hashing techniques for classification in federated learning, which reduces the output size of the model and improves the quality of the hash codes. By using a binary cross-entropy-based quantization loss, FedHAP achieves significant improvements in accuracy, communication cost, and convergence rate.

### C. Classification Performance

We evaluated the identification performance using three widely adopted metrics: classification accuracy (ACC), F1-score, and the area under the receiver operating characteristic curve (AUC). Higher values for these metrics indicate better performance. The global model is evaluated uniformly across all centers. In contrast, for personalized federated learning methods, each site's model is tested on its local data to ensure a fair comparison under heterogeneous distributions. We reported the mean and standard deviation of these metrics for each method to reflect both overall performance and stability.

As shown in Table I, our method outperforms the comparison methods in most of the metrics. For example, our proposed method achieves the highest accuracy of 78.78% on the ABIDE dataset, outperforming the second-best method (CusFed) by 3.66%. Similar trends are observed on the

TABLE II  
ACCURACY (%) OF ABLATION EXPERIMENTS ON ABIDE, ADHD-200, AND SCHIZOPHRENIA DATASETS. FHP: FEDERATED HASH PROTOTYPE. DMA: DYNAMIC MODEL AGGREGATION. HE: HASH ENCODING.

| FHP | DMA | HE | ABIDE      |            |            | ADHD-200   |            |            | Schizophrenia |            |            |
|-----|-----|----|------------|------------|------------|------------|------------|------------|---------------|------------|------------|
|     |     |    | ACC        | F1         | AUC        | ACC        | F1         | AUC        | ACC           | F1         | AUC        |
| ✓   | ×   | ×  | 65.51±0.08 | 59.72±0.32 | 58.54±0.08 | 66.51±0.09 | 51.69±0.05 | 54.80±0.04 | 65.80±0.08    | 43.15±0.11 | 61.13±0.02 |
| ×   | ✓   | ×  | 69.00±0.06 | 66.55±0.14 | 64.88±0.07 | 67.70±0.22 | 57.65±0.38 | 62.44±0.06 | 68.19±0.05    | 47.96±0.09 | 57.64±0.06 |
| ×   | ×   | ✓  | 67.61±0.56 | 59.29±0.18 | 65.11±0.21 | 74.63±0.13 | 51.17±1.36 | 62.78±0.03 | 68.46±0.16    | 55.75±0.35 | 63.55±0.06 |
| ✓   | ✓   | ×  | 70.64±0.11 | 67.16±0.07 | 66.64±0.07 | 73.85±0.08 | 50.75±0.42 | 62.93±0.03 | 71.79±0.14    | 41.57±0.12 | 62.39±0.07 |
| ×   | ✓   | ✓  | 69.98±0.01 | 60.88±0.12 | 65.82±0.06 | 73.33±0.23 | 69.04±0.12 | 71.67±0.25 | 73.25±0.05    | 52.45±1.31 | 62.95±0.04 |
| ✓   | ×   | ✓  | 70.25±0.10 | 60.12±0.00 | 64.98±0.12 | 73.71±0.33 | 67.61±0.45 | 73.11±0.32 | 74.47±0.08    | 59.41±0.08 | 70.38±0.07 |
| ✓   | ✓   | ✓  | 78.78±0.09 | 62.22±0.21 | 74.58±0.08 | 76.84±0.31 | 65.77±0.13 | 75.32±0.29 | 77.77±0.34    | 71.18±0.52 | 73.31±0.32 |

ADHD-200 and SZ datasets, where our method achieves 76.84% and 77.77% accuracy, respectively. These results demonstrate that the effectiveness of our method in multi-site brain network analysis. Moreover, we computed the standard deviation of these metrics for each method. Furthermore, our framework yields notable improvements in AUC, achieving 74.58%, 75.32%, and 73.31% on ABIDE, ADHD-200, and SZ, respectively, outperforming prior federated and prototype-based approaches. The AUC gains indicate that the proposed hash-prototype-guided feature space effectively separates patients from healthy controls.

To further validate the statistical reliability of the performance improvements, we conducted paired statistical significance tests between the proposed method and the strongest competing baseline on each dataset. Specifically, the proposed method was compared with CusFed on ABIDE, FedALA on ADHD-200, and FedHAP on Schizophrenia. The corresponding p-values are reported in Table I. The results indicate that the proposed method achieves statistically significant improvements over the strongest competing baselines on the main evaluation metrics, further demonstrating the effectiveness and robustness of the proposed hash-prototype-guided personalized federated learning framework.

## V. DISCUSSION

### A. Ablation Study

In the ablation study, we systematically explored the contributions of three key components in our proposed personalized federated learning method: dynamic model aggregation, hash encoding, and federated hash prototype. Through an exhaustive combination of these factors, we observed that their joint utilization resulted in the most significant performance enhancement. Specifically, when omitting the hash encoding layer, we replaced it with a single fully connected layer in order to discern the isolated impact of the hash encoding process. Furthermore, to evaluate the significance of dynamic model aggregation, we performed experiments where personalized aggregation was replaced with a simple averaging operation. Remarkably, in both scenarios, the absence of either hash encoding or dynamic model aggregation led to a notable degradation in performance, underscoring the essential contributions of these components.

To provide a comprehensive evaluation of our method, we conducted experiments on three diverse datasets, comparing the performance metrics of ACC, F1-score, and AUC. As shown in Table II, our model consistently outperformed state-of-the-art methods across all datasets. These results not only validate the effectiveness of our proposed method but also indicate its robustness and generalizability across different domains, further emphasizing its potential for practical applications. The combination of these elements contributes to the superior performance observed in our experimental evaluations, demonstrating the method's effectiveness in both accuracy and stability.

We conducted comparative experiments between the GAT and GCN models on ABIDE, ADHD-200, and SZ datasets. Traditional GCN methods typically rely on a static, pre-defined adjacency matrix to aggregate neighbor information based on fixed weights. However, brain functional connectivity varies significantly between subjects. GAT allows the input-dependent edge weighting, enabling the model to highlight patient-specific functional abnormalities. As shown in Fig. 3, GAT achieves an ACC of 78.8% and an F1-score of 76.5% on the ABIDE dataset. This is a substantial improvement compared to the GCN method, which only scores 62.0% and 60.1%. We observe consistent improvements on the ADHD-200 and SZ datasets. These results further suggest the effectiveness of the GAT for modeling heterogeneous and subject-specific brain networks. Such adaptive representations are particularly effective for capturing individual variability in psychiatric disorder classification.

### B. Convergence Analysis

We conducted a comparative analysis of convergence efficiency, a crucial criterion for evaluating the effectiveness of federated learning methods. As shown in Fig. 4, we present the average accuracy on three brain disease datasets. The number of local learning epochs between multi-site communications was set to 4 and 8, respectively. The experiment results show that compared to the state-of-the-art methods, our approach demonstrates a significant advantage in convergence speed. These results suggest that our federated aggregation scheme effectively integrates global information into each site's model, enhancing local model performance.

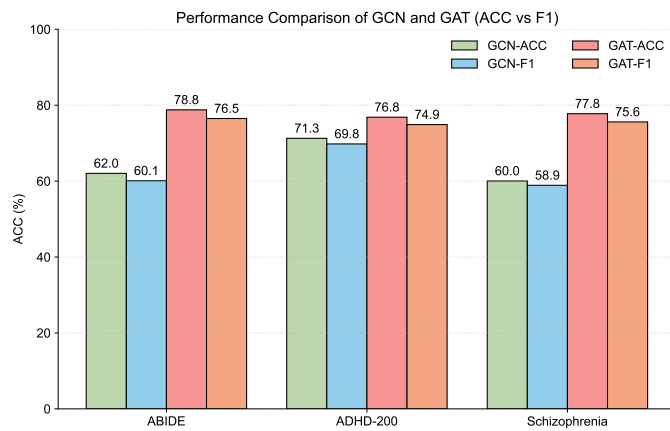


Fig. 3. Experimental results of our method and the GCN method on ABIDE, ADHD-200, and Schizophrenia datasets.

### C. Feature Alignment Analysis

The utilization of hash prototypes in our federated learning framework enables a balance between global consistency and local specificity. During the training phase, the prototypes are aggregated and disseminated among participating sites, enabling the generation of new prototypes that guide the subsequent model training iterations. The aggregation mechanism ensures that each site benefits from the collective knowledge while maintaining the personalized aspects of its own dataset. To evaluate whether the hash prototypes can effectively align with data distributions across multiple centers, we perform t-SNE visualization on the hash-coded features. Specifically, for each dataset at each center, 10 healthy subjects and 10 patients are randomly selected for analysis. It should be noted that only correctly classified samples are visualized to evaluate the effectiveness of the hash prototypes. Similarly, the hash topology prototype is also visualized (in triangles) in this section. As shown in Fig. 5, the hash topology prototype learned by the proposed model aligns well with the data at each center. The model achieves both inter-center generalization and intra-center personalization, which are essential for modeling heterogeneous medical data.

### D. Brain Network Biomarkers

Furthermore, our method can more effectively identify multi-center brain network biomarkers due to the learned prototypes. The proposed method not only evaluates feature importance but also reveals interaction patterns and organizational structures among brain regions, offering insights into their roles and mutual influences in the development of brain disorders. Fig. 6 illustrates the spatial distribution of discriminative brain regions identified across multiple large-scale neuroimaging datasets, including ABIDE, ADHD-200, and SZ. Red lines represent positive correlations between brain regions, whereas green lines represent negative correlations. The circular diagram visualizes the intricate functional network interactions among these regions. Notably, the highlighted regions exhibit consistent overlap with functional networks implicated in the respective disorders. For the ABIDE dataset, the discriminative regions were mainly distributed in the

temporal cortex, parahippocampal regions, anterior cingulate cortex, and default-mode-network-related regions, which are strongly associated with social cognition and self-referential processing deficits characteristic of autism spectrum disorder. According to the recurrence frequency, the most representative regions included Temporal\_Sup, Frontal\_Inf\_Orb, and Parahippocampal. These results suggest that the identified biomarkers are not merely site-specific visual patterns but show a degree of stability across sites.

For the ADHD-200 dataset, the discriminative regions are predominantly located in the prefrontal and parietal cortices, reflecting impairments in attentional control and executive functioning, which are associated with attention-deficit/hyperactivity disorder. In contrast, the SZ dataset highlights abnormalities within the fronto-parietal and fronto-temporal circuits, including the prefrontal cortex and hippocampal regions, aligning with dysconnectivity theories of SZ that emphasize impaired cognitive control and memory dysfunction. Collectively, these regional distributions provide biologically meaningful biomarkers.

### E. Limitations and Future Work

Although the proposed method obtains superior performance in multi-site brain network analysis, the limitations and future directions are summarized as follows. First, we only use resting-state fMRI (rs-fMRI) data to construct brain networks in this method. However, brain diseases are often manifested through complex structural and functional changes. Therefore, we will extend the shared encoder to extract multi-modal data, such as combining functional connectivity with structural MRI (sMRI) or diffusion tensor imaging (DTI), to enhance the model's capacity and improve its robustness. Second, the proposed framework is evaluated only on binary classification tasks using fMRI data, and its effectiveness on multi-class problems or other neuroimaging modalities has not been explored. In future studies, we will replace the binary cross-entropy loss with categorical cross-entropy and utilize more balanced datasets. Third, although the proposed hash encoding mechanism reduces information exposure by transmitting compact binary prototypes instead of raw neuroimaging data, it lacks the formal privacy guarantees of cryptographic methods. The shared hash prototypes and encoder parameters may still contain statistical information about local data distributions and may be vulnerable to advanced inference attacks, such as membership inference or prototype reconstruction attacks [43]. Future work will investigate the integration of differential privacy, secure aggregation, and privacy attack evaluation protocols to further strengthen the security of the proposed framework. Such efforts are expected to further enhance the generalization capability and practical applicability.

## VI. CONCLUSION

In conclusion, we propose a hash prototype-based personalized federated learning method for multi-site brain network analysis. Our method effectively addresses the critical challenges of data privacy and model personalization in collaborative diagnostic settings. Firstly, we decouple local

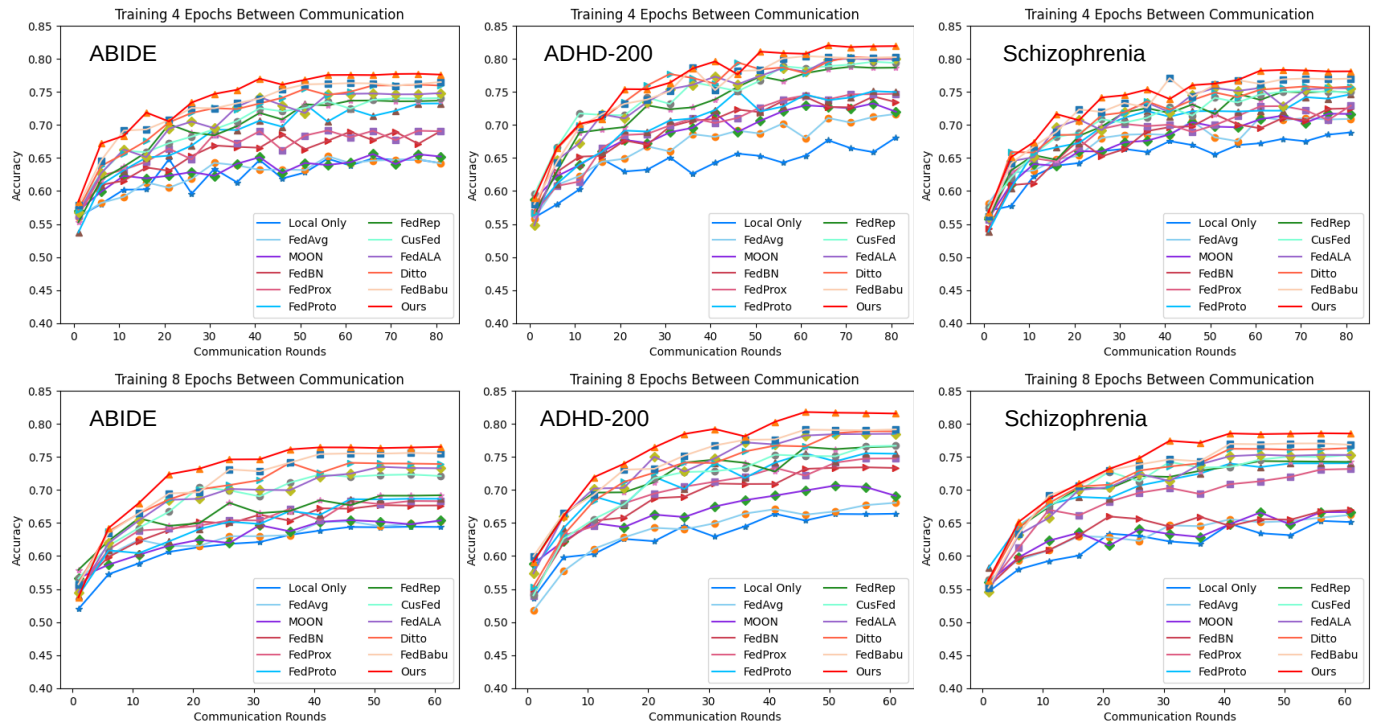


Fig. 4. Convergence analysis of different methods.

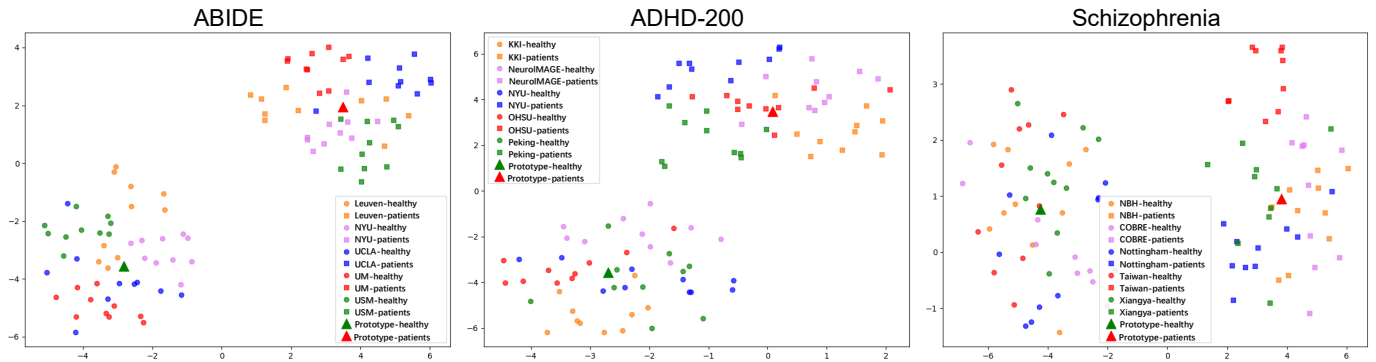


Fig. 5. Visualization of hash prototypes for multi-center data alignment by our method.

models into shared encoders to preserve site-specific features and a hash encoding layer to generate compact binary representations with reduced information exposure. Secondly, these hash prototypes are aggregated and shared across sites by generating local hash prototypes, supporting cross-site alignment of data distributions and guiding model training. Furthermore, encoder parameters are dynamically aggregated to integrate global knowledge with local specificity while maintaining personalized feature extraction for each site. Finally, we employ a multi-layer perceptron approach to classify the extracted features for multi-site brain network analysis. Extensive experiments on multi-site brain disease datasets demonstrate consistent improvements over the state-of-the-art federated learning methods across accuracy, F1-score, and other standard evaluation metrics. Moreover, our proposed method can enhance diagnostic performance without sharing raw data, while simultaneously identifying both cross-site and

site-specific biomarkers, providing valuable insights for the analysis of neuropsychiatric disorders.

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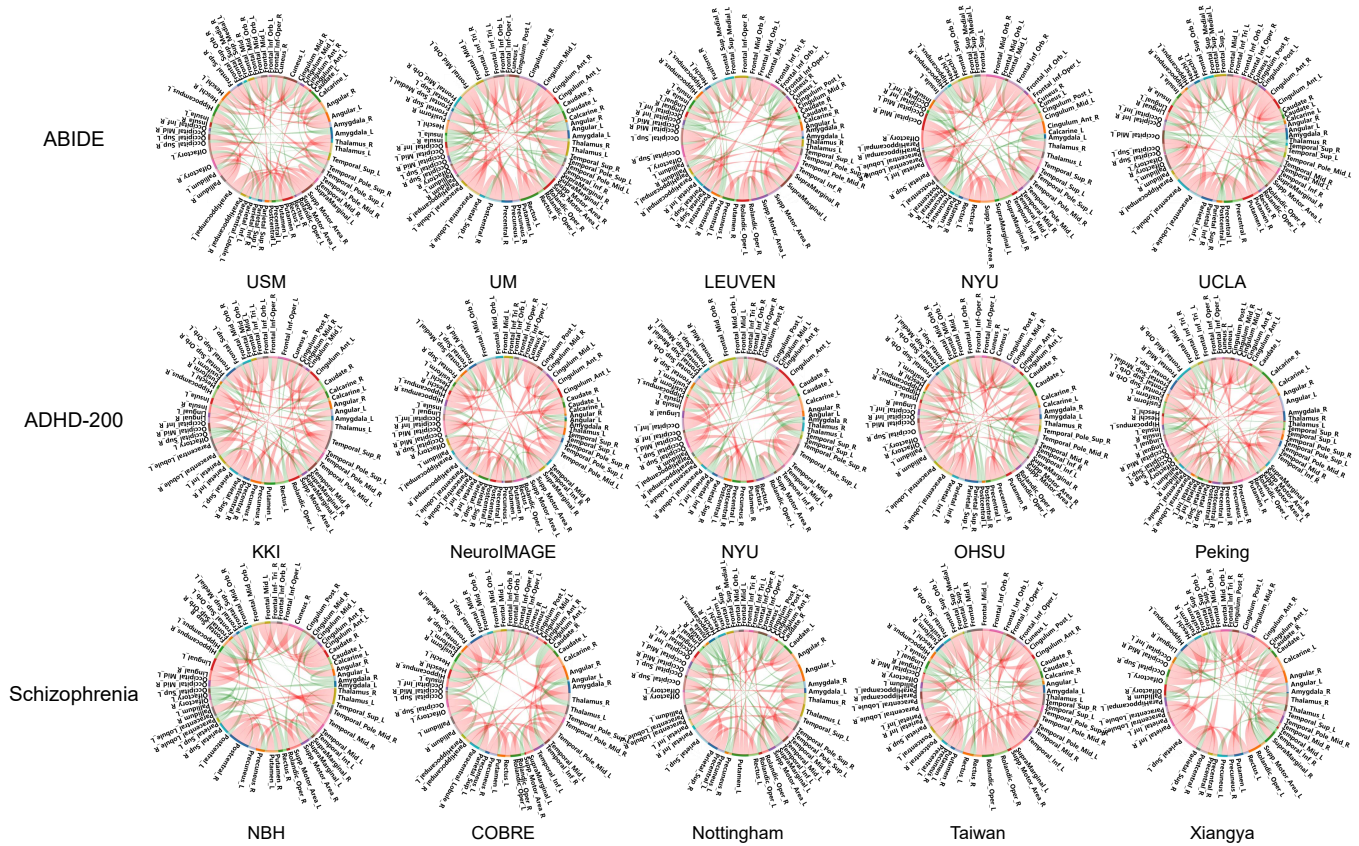


Fig. 6. Spatial distribution of discriminative brain regions across multiple datasets.

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